

Finite Symbolic Perturbations in Chained Language Handlers

A Research Note on Nonlinear Dynamical Language Filtering, Trajectory Steering, and
Attractor Formation

September 2026

Pensée

Abstract

This research note develops a simple experimental observation into a framework for studying chained interactions between large language models as a nonlinear dynamical language network. Three otherwise identical prompts concerning football were supplied to DeepSeek, differing only in the appended symbolic markers `[good]`, `[bad]`, and `[neutral]`. The resulting outputs followed conspicuously different semantic trajectories: positive social and cultural effects, problems and criticisms, and comparatively descriptive or encyclopaedic exposition. The experiment does not by itself establish the existence of a literal latent manifold or dynamical attractor. It does, however, demonstrate an operationally useful phenomenon: a very small finite symbolic perturbation can redirect a much larger generated language trajectory. This observation is connected here to Functional Symbolic Trajectories (FSTs), Functional Symbolic Dynamics (FSD), the Dynamic Semantic Lattice, nodal valuation, projection-cone interactions, and an existing model in which chained language handlers act as nonlinear filters. The note proposes that small valuation markers can serve as controlled probe and steering signals for investigating trajectory propagation, bifurcation, persistence, convergence, and possible attractor-like behaviour across networks of heterogeneous language handlers.

1 Introduction

A language-model interaction is commonly described in terms of prompts and responses. For the present purpose, a different description may be more useful. A prompt can be treated as a finite symbolic input trajectory presented to a language handler, while the generated response is another finite symbolic trajectory produced under the constraints of that handler. The handler need not initially be interpreted as a reasoning agent, nor is it necessary to make claims concerning an inaccessible internal semantic reality. Operationally, it can be treated as a nonlinear dynamical language filter whose observable behaviour is expressed through changes in symbolic trajectories.

This framing becomes particularly useful when language handlers are chained. The output of one handler becomes the input of another, so that the second handler does not receive some unsymbolised original meaning. It receives the finite symbolic trajectory produced by the preceding handler. A chain of language models can therefore be considered as a network through which finite symbolic trajectories are repeatedly projected, filtered, reconstructed, redirected, and re-expressed.

A small experiment provides a useful probe of this process. The same request concerning football was presented three times, with only a final valuation marker changed. The prompts were of the form

$$\begin{aligned}P_+ &= \text{“Please tell me about football [good].”}, \\P_- &= \text{“Please tell me about football [bad].”}, \\P_0 &= \text{“Please tell me about football [neutral].”}.\end{aligned}$$

Despite the very small symbolic difference between these inputs, the resulting responses developed along substantially different semantic directions. This suggests that such markers can be used not merely as labels but as experimental perturbations through which the trajectory-forming behaviour of language handlers can be examined.

2 The Three-Prompt Observation

The [good] prompt generated a response organised around football as a positive social phenomenon. Its trajectory moved through themes including football as a global language, cultural connection, historical development, health and wellbeing, social inclusion, community, confidence, teamwork, and leadership.

The [bad] prompt generated a markedly different trajectory. The response concentrated upon VAR and refereeing, money and commercialisation, ticket prices, subscription costs, player welfare, fixture congestion, governance, punditry, and concerns about the “soul of the game.”

The `[neutral]` prompt instead produced a comparatively encyclopaedic trajectory. It described the objective of the game, fundamental rules, match duration, restarts, historical origins, major institutional milestones, player positions, and formations.

The result is modest but useful. The three prompts differ by only one small terminal symbolic component, while the generated responses differ in their organising direction. The appended term appears therefore to act as more than an additional subject word. It supplies an orientation under which the larger subject is traversed.

If the broad semantic possibilities associated with football are provisionally represented by a region $\mathcal{M}_F$, then a simple first approximation would be

$$P_+ \longrightarrow \mathcal{M}_F \cap \mathcal{M}_+, \quad (1)$$

$$P_- \longrightarrow \mathcal{M}_F \cap \mathcal{M}_-, \quad (2)$$

$$P_0 \longrightarrow \mathcal{M}_F \cap \mathcal{M}_0. \quad (3)$$

This notation should not be read as establishing that the model literally contains such sets. It is a functional description of the observed partitioning of generated trajectories. More importantly, the outputs suggest not merely different collections of facts but different directions of development. The marker appears to influence the trajectory from its earliest generated stages.

3 Finite Symbolic Perturbation

Let T denote an input symbolic trajectory and let H denote a language handler. The handler produces

$$T' = H(T). \quad (4)$$

Now introduce a small finite symbolic perturbation δT . In the football experiment this perturbation may be as small as replacing `[good]` with `[bad]`. The observed behaviour suggests that, in general,

$$H(T + \delta T) \neq H(T) + H(\delta T). \quad (5)$$

The expression is not intended as a literal algebra of natural-language strings. It records the relevant functional observation: the effect of the added symbol depends upon the larger trajectory in which it occurs, and a small input alteration may reorganise a much larger output.

This gives a useful experimental form:

$$\boxed{\text{small finite symbolic perturbation} \longrightarrow \text{large trajectory divergence.}} \quad (6)$$

The information subsequently expressed is clearly not contained in the word **good** itself. Rather, the marker appears to select, weight, or steer relations already accessible to the handler. A small symbolic term can therefore have a disproportionately large effect upon the path subsequently taken through available language relations.

4 FST Initialisation and Orientation

Within Functional Symbolic Dynamics, the experiment can be interpreted as an initialisation problem. Let an interaction be initialised by a subject S and an orientation or valuation V :

$$I = [S, V]. \quad (7)$$

For the present experiment,

$$\begin{aligned} I_+ &= [\text{football}, \text{good}], \\ I_- &= [\text{football}, \text{bad}], \\ I_0 &= [\text{football}, \text{neutral}]. \end{aligned}$$

Each initialisation gives rise to a different Functional Symbolic Trajectory:

$$\text{football} \begin{cases} +[\text{good}] & \longrightarrow T_+, \\ +[\text{neutral}] & \longrightarrow T_0, \\ +[\text{bad}] & \longrightarrow T_-. \end{cases} \quad (8)$$

This provides a useful analogy with hypothesis formation. A hypothesis need not be regarded as a permanent commitment to an external ontology. It may instead serve as a temporary FST initialisation: a finite symbolic condition that constrains which trajectories are explored next. The experiment suggests that even a very small initialisation term may be sufficient to produce a substantial change in the subsequent symbolic path.

5 The Handler as a Nonlinear Dynamical Language Filter

The observation fits naturally into the chained-handler model. Let H_n denote the n th language handler and T_n the finite symbolic trajectory entering it. Then

$$T_{n+1} = H_n(T_n). \quad (9)$$

For a chain of N handlers,

$$T_N = H_N \circ H_{N-1} \circ \cdots \circ H_2 \circ H_1(T_0). \quad (10)$$

The significance of this expression is conceptual. Handler H_2 does not encounter T_0 independently. It encounters $T_1 = H_1(T_0)$. Similarly, H_3 encounters a trajectory already transformed by both preceding handlers. The chain therefore has history. Its present input contains the symbolic consequences of previous interactions.

This differs from treating several language models simply as independent question-answering systems. In a chain, every handler operates upon an inherited symbolic state. The output of one language geometry becomes a perturbation presented to another.

If valuation or steering symbols are explicitly introduced, the expression may be extended to

$$T_{n+1} = H_n(T_n, V_n), \quad (11)$$

where V_n is a finite symbolic steering condition. A chain can then be represented as

$$T_0 \rightarrow H_1 \rightarrow T_1 \rightarrow V_1 \rightarrow H_2 \rightarrow T_2 \rightarrow V_2 \rightarrow H_3 \rightarrow T_3. \quad (12)$$

The valuation term need not contain the semantic richness of the subsequent output. Its role may instead be to alter the direction from which the handler reconstructs and develops the incoming trajectory.

6 Dynamic Semantic Lattice and Nodal Steering

The same phenomenon can be expressed within the Dynamic Semantic Lattice model. If generated language is treated as a traversal through dynamically activated semantic relations, a marker such as [good] can be interpreted as a low-dimensional nodal value whose effect propagates through subsequent traversal.

A minimal functional representation is

$$T_{t+1} = F(T_t, S, V), \quad (13)$$

where T_t is the existing trajectory, S is the subject constraint, and V is the orientating value. The three football trials then correspond approximately to

$$F(T_0, \text{football}, +) \longrightarrow T_+, \quad (14)$$

$$F(T_0, \text{football}, -) \longrightarrow T_-, \quad (15)$$

$$F(T_0, \text{football}, 0) \longrightarrow T_0. \quad (16)$$

This formulation suggests a distinction between a *nodal value* and a *trajectory value*. A valuation may apply to a particular symbol or node,

$$V(x_n), \quad (17)$$

but it may also be applied retrospectively to a complete path,

$$V(x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_n). \quad (18)$$

The second possibility is especially important for chained interactions. Two trajectories may terminate in similar linguistic regions while having arrived there through different routes. Valuing only the endpoint loses that distinction; valuing the FST preserves it.

7 Projection Cones Between Language Handlers

A complementary geometric picture is supplied by the projection-cone model. Each language handler may be provisionally represented as possessing a very large space of possible language relations. During an actual interaction only a restricted region is activated or made relevant by the incoming finite symbolic trajectory.

The incoming text can therefore be pictured as defining a projection cone onto a restricted interaction region. A steering term such as [good] does not necessarily enlarge the cone by supplying much additional subject information. Instead, it may rotate, incline, narrow, or otherwise redirect the projection.

In a heterogeneous chain,

$$H_A \rightarrow H_B \rightarrow H_C, \quad (19)$$

where the handlers may be different models, the observable sequence can be written as

$$\mathcal{S}_A \xrightarrow{T_1} \mathcal{S}_B \xrightarrow{T_2} \mathcal{S}_C, \quad (20)$$

where $\mathcal{S}_A$, $\mathcal{S}_B$, and $\mathcal{S}_C$ denote the provisionally conceived language spaces of the handlers. What

passes between them is not the whole space but a finite symbolic trajectory.

Thus the operational cycle is

$$\boxed{\text{large language possibility}} \rightarrow \boxed{\text{finite symbolic trajectory}} \rightarrow \boxed{\text{large language possibility}} \rightarrow \boxed{\text{finite symbolic trajectory}} \quad (21)$$

This formulation is deliberately conservative. The finite trajectory is observable. The internal manifold is inferred. The experimental programme can therefore concentrate upon perturbing and measuring trajectories without requiring ontological commitment to a particular internal geometry.

8 Branching, Bifurcation, and Recombination

The chained model naturally extends to approximately parallel or bifurcating arrangements. An initial trajectory may be presented independently to several handlers:

$$T_0 \begin{cases} \rightarrow H_A \rightarrow T_A, \\ \rightarrow H_B \rightarrow T_B, \\ \rightarrow H_C \rightarrow T_C. \end{cases} \quad (22)$$

The resulting trajectories may then be recombined through a further handler,

$$(T_A, T_B, T_C) \rightarrow H_D \rightarrow T_D. \quad (23)$$

This is better described as symbolic bifurcation and recombination than as simple classical parallel processing. Each branch develops according to the nonlinear filtering properties of its handler, and the recombination stage receives the finite symbolic products of those separate developments.

Steering markers provide a particularly simple means of probing such a network. The same T_0 can be supplied with different finite perturbations, and the propagation of the resulting divergence can be followed across branches and recombinations.

9 From Steering to Attractor-Like Behaviour

The initial football experiment should not be taken as sufficient evidence that the marker [good] creates an attractor. It demonstrates something narrower and more secure: a small symbolic change can initialise a substantially different output trajectory.

It is useful to distinguish three phenomena:

$$\boxed{\text{lexical steering}} \neq \boxed{\text{persistent interaction state}} \neq \boxed{\text{learned or recurrent attractor}}. \quad (24)$$

The present observation strongly motivates investigation of the first. A second stage would test whether an orientation persists after its explicit marker has disappeared. A third would test whether repeated or chained trajectories converge toward characteristic regions despite variation in their initial conditions.

In this sense, the term *attractor* becomes experimentally useful only when recurrence, convergence, or persistence can be demonstrated. A manifold may likewise remain a provisional reconstruction rather than an assumed internal object. Functional Symbolic Dynamics permits the investigation to proceed from finite observable trajectories while stronger geometric interpretations remain hypotheses.

10 A Proposed Experimental Programme

A simple controlled programme follows directly from the observation. Begin with a reference trajectory T and produce minimally perturbed variants

$$T^+ = T + [\text{good}], \quad T^- = T + [\text{bad}], \quad T^0 = T + [\text{neutral}]. \quad (25)$$

Pass all three through the same handler H_A :

$$T_A^+ = H_A(T^+), \quad T_A^- = H_A(T^-), \quad T_A^0 = H_A(T^0). \quad (26)$$

The outputs can then be passed independently to a second handler:

$$T_B^+ = H_B(T_A^+), \quad T_B^- = H_B(T_A^-), \quad T_B^0 = H_B(T_A^0), \quad (27)$$

and continued through additional handlers. The experiment then asks whether the original finite perturbation grows, contracts, disappears, reverses, bifurcates, or stabilises as it propagates through the network.

A conceptual divergence variable may be written

$$\delta_n = D(T_n^+, T_n^-), \quad (28)$$

where D is some explicitly defined finite comparison between trajectories. No particular metric

is assumed here. Possible measures could later include lexical divergence, topic-path divergence, embedding separation, trajectory curvature, recurrence of semantic nodes, or a specifically FSD-derived finite symbolic measure.

A second experiment would place the valuation marker after a substantial interaction:

$$P_1 \rightarrow R_1 \rightarrow P_2 \rightarrow R_2 \rightarrow P_3 \rightarrow R_3 \rightarrow [\text{good}], \quad (29)$$

followed by an ambiguous continuation such as “Continue.” Matched trials ending in `[bad]` and `[neutral]` would test whether retrospective valuation of an established trajectory alters subsequent persistence.

A third experiment would remove both the subject and the valuation marker from the continuation. After a long discussion ending in `[good]`, the handler might receive only “Tell me something else about it.” Persistence of the previously established orientation would provide stronger evidence for an interaction state extending beyond direct lexical triggering.

Finally, repeated chains with controlled perturbations could test convergence. If substantially different initial trajectories repeatedly approach similar symbolic regions after passage through the same handler network, an attractor-like description would acquire operational support.

11 A Minimal Network Abstraction

The emerging architecture can be compressed into three principal components:

$$\boxed{\text{FSD Language Network} = \text{Handlers} + \text{FSTs} + \text{Steering Symbols.}} \quad (30)$$

The *handler* is the nonlinear language filter. The *FST* is the finite observable trajectory propagated between handlers. The *steering symbol* is a finite perturbation capable of changing subsequent trajectory formation.

This abstraction has an important methodological advantage. It does not require the experimenter to know what the handler “really means,” whether its latent representation is literally manifold-like, or whether a generated trajectory corresponds to an internal symbolic object. The network can be investigated through controlled finite inputs and finite observable outputs.

A suitably narrow empirical proposition is therefore:

A finite symbolic perturbation applied to an input trajectory can produce a reproducible change in the output trajectory of a language handler, and that changed trajectory can itself be propagated through a network of subsequent language handlers.

The first part is suggested by the three football trials. The second is directly testable through chained interaction experiments.

12 Discussion

The simplicity of the football experiment is an advantage. The markers [good], [bad], and [neutral] contain little domain information, yet they produce conspicuously different expositions of the same domain. This makes them useful as probe signals. They allow a controlled perturbation to be introduced without replacing the original subject with a heavily specified instruction.

The experiment also suggests an important distinction between information and direction. The marker does not contain the historical, social, financial, regulatory, or tactical material subsequently generated. Its apparent role is directional. In the projection-cone picture it alters the orientation of the interaction region. In the Dynamic Semantic Lattice it supplies a nodal steering value. In FSD it modifies the initial condition from which the next FST develops. In the chained-handler model it becomes a small control signal whose consequences can be followed through a nonlinear network.

There are, however, several open questions. A single model response may reflect ordinary instruction following rather than a persistent dynamical state. The apparent semantic regions may be artifacts of the observer's chosen description. Different sampling parameters or system instructions may change the scale of divergence. Repeated trials are therefore required before stronger claims concerning stability or attractors are made. These are not objections to the experimental framing; they define the measurements that the framing makes possible.

The deeper opportunity lies in moving away from asking only whether two models agree, disagree, reason correctly, or produce similar answers. A chained FSD experiment instead asks how a controlled perturbation propagates through successive nonlinear symbolic transformations. Does it decay? Does it amplify? Does one handler erase a distinction introduced by another? Do heterogeneous handlers stabilise particular trajectories? Can bifurcated trajectories reconverge? Can a small valuation applied retrospectively alter the future path of an established interaction? These are questions about dynamics rather than merely content.

13 Conclusion

Three short prompts provide a useful starting point. Changing only [good], [bad], and [neutral] produced three substantially different trajectories through the broad subject of football. The observation does not establish a hidden manifold or attractor, but it does show how a tiny finite symbolic perturbation can act as a strong trajectory-selection condition.

Placed within the wider framework of Functional Symbolic Dynamics, this suggests a practical

model of chained language interaction. Language handlers can be treated as nonlinear filters; their outputs as Functional Symbolic Trajectories; and small valuation markers as steering signals. Chaining handlers then produces a nonlinear dynamical language network in which symbolic trajectories may propagate, bifurcate, recombine, stabilise, or diverge.

The important methodological point is that the observable trajectory remains primary. Manifolds, basins, and attractors can be introduced progressively as reconstructed descriptions if repeated experiments justify them. In this way the framework permits dynamical questions to be asked of language systems without requiring prior commitment to an inaccessible internal ontology.

The small marker at the end of a prompt may therefore be more than a label. It may serve as an experimental handle upon the direction of a language trajectory—a finite symbolic perturbation through which the larger dynamics of chained language handlers can begin to be mapped.